EXPERIMENT - 14

AIM:

To write a code using RAW sockets to implement Packet sniffing

CODE:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def Packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        Protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        Protocol_name = ""
        if Protocol == 1:
            Protocol_name = "ICMP"
        elif Protocol == 6:
            Protocol_name = "TCP"
        elif Protocol == 17:
            Protocol_name = "UDP"
        else:
            Protocol_name = "unknown Protocol"
        Print(f"Protocol: {Protocol_name}")
        Print(f"source IP: {src_ip}")
        Print(f"Destination IP: {dst_ip}")
        Print("-"*50)
Sniff(iface = "wiFi", Prn = Packet_callback,
      filter = "ip", store = 0)
```

Scanned with OKEN Scanner

OUTPUT

Protocol : TCP
Source IP: 57.155.141.117
Destination IP : 192.168.1.6

Protocol : UDP
Source IP : 57.155.141.117
Destination IP : 192.168.1.1

Protocol : UDP
Source IP : 57.155.141.117
Destination IP : 192.168.1.1

Protocol : UDP
Source IP : 57.155.141.117
Destination IP : 192.168.1.1

RESULT: Thus the Program code using RAW sockets to implement Packet sniffing was executed successfully